Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Projectiles

MARKSCHEME ($g = 9.8\,\text{m}\,\text{s}^{-2}$)

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1. $u_x = 10\sqrt{3}$, $u_y = 10$ **M1A1** [3]
2. $T = \frac{2u\sin\alpha}{g}$; $R = u\cos\alpha \cdot T$ **M1A1** [5]
3. $H = \frac{u^2\sin^2\alpha}{2g}$; $R = \frac{u^2\sin 2\alpha}{g}$ **M1A1** [5]
4. $t = \sqrt{\frac{2h}{g}}$; $x = ut$ **M1A1** [5]
5. $v_x = u\cos\alpha$; $v_y = u\sin\alpha - gt$; speed $\sqrt{v_x^2 + v_y^2}$ **M1A1** [5]
6. Eliminate t: $y = x\tan\alpha - \frac{gx^2}{2u^2\cos^2\alpha}$ **M1A1** [4]
7. $x = (u\cos\alpha)t$; $y = (u\sin\alpha)t - \frac{1}{2}gt^2$ **M1A1** [4]
8. $t = \sqrt{\frac{2h}{g}}$; range $= ut$ **M1A1** [5]
9. $H = \frac{u^2\sin^2\alpha}{2g}$ **M1A1** [3]
10. $R = \frac{u^2\sin 2\alpha}{g}$ **M1A1** [4]
11. $y = 5 = (u\sin\alpha)t - \frac{1}{2}gt^2$; solve quadratic **M1A1** [5]
12. $0 = (u\sin\alpha)T - \frac{1}{2}gT^2 \Rightarrow T = \frac{2u\sin\alpha}{g}$ **M1A1** [3]
13. $R = (u\cos\alpha)\frac{2u\sin\alpha}{g} = \frac{u^2\sin 2\alpha}{g}$ **M1A1** [3]
14. $R = \frac{u^2\sin 2\alpha}{g}$ with $\sin 106^\circ$ **M1A1** [4]
15. $v_y = -\sqrt{2gh}$; speed $\sqrt{u^2 + 2gh}$ **M1A1** [5]
16. Quadratic for t from $y = -20$; then $x = u_x t$ **M1A1** [7]
17. $10 = (u\sin\alpha)t - \frac{1}{2}gt^2$; two roots; corresponding x **M1A1** [6]
18. $\sin\alpha = \frac{3}{5}, \cos\alpha = \frac{4}{5}$; R, H formulae **M1A1** [5]
19. Time to wall $= \frac{50}{u\cos\alpha}$; compare height with 0 **M1A1** [6]
20. $\frac{dR}{d\alpha} = 0 \Rightarrow \alpha = 45^\circ$; $R_{\max} = \frac{u^2}{g}$ **M1A1** [4]
21. Impact: energy or components; direction $\tan^{-1}(v_y/v_x)$ **M1A1** [6]
22. $y = \frac{3}{2}x - \frac{9.8x^2}{200}$; set $y = 0$ for range **M1A1** [5]
23. $a_{\parallel} = -g\sin\beta$; $a_{\perp} = -g\cos\beta$ **M1A1** [4]
24. Resolve axes along/perp plane; range formula on incline **M1A1** [7]
25. Equate positions; solve simultaneous for t **M1A1** [7]